15. Karnal Bunt is a disease of :

(a) Barley crop (b) Wheat crop

(c) Bajra crop (d) Jowar crop

U.P. R.O./A.R.O. (Pre) 2016

Ans. (b)

Karnal Bunt is a fungal disease caused by the smut fungus Tilletia indica. It affects the quality of wheat. It was first reported in 1931, infecting wheat growing near the city of Karnal, Haryana.

16. Tungro virus of rice is spread by –

(a) Borer (b) Inflorescent Bug

(c) Golmiz (d) Green leafhoppers

U.P.P.C.S. (Mains) 2009

Ans. (d)

Tungro is a disease of rice found in South-East Asia. The cause of disease are two group of virus :
(1) RTSV : Rice Tungro Spherical Virus
(2) RTBV : Rice Tungro Bacilliform Virus
Both are transmitted together by green leafhoppers.

17. Consider the following :

1. Birds 2. Dust blowing

3. Rain 4. Wind blowing

Which of the above spread plant diseases?

(a) 1 and 3 only (b) 3 and 4 only

(c) 1, 2 and 4 only (d) 1, 2 , 3 and 4

I.A.S. (Pre) 2018

Ans. (d)

Plant Pathology (also Phytopathology) is the scientific study of plant diseases caused by pathogens like viruses, bacteria, fungi etc. and environmental conditions (physiological factors). The dissemination factors of these include – wind, rain, animals, soil, nursery graft, contaminated equipment and tools, infected seed stock, pollen, dust storm, irrigation water, birds and humans.

18. What causes 'Blackheart' in potato ?

(a) Copper deficiency (b) Boron deficiency

(c) Oxygen deficiency (d) Potassium deficiency

U.P.P.C.S. (Spl.) (Pre) 2008

Ans. (c)

Blackheart is an abiotic disease which is caused due to low availability of oxygen during storage of potato.

19. Yellow vein mosaic disease in okra, caused by :

(a) Aphids (b) Whitefly

(c) Leafhopper (d) Fungi

U.P.P.C.S. (Mains) 2002

Ans. (b)

Yellow vein mosaic disease in okra is a viral disease. Whitefly is a factor of this virus.

20. 'Yellow Vein Mosaic' is a serious disease of :

(a) Brinjal (b) Okra

(c) Pea (d) Cabbage

U.P. P.C.S. (Mains) 2016

Ans. (b)

See the explanation of the above question.

21. 'Spongy Tissue' is a serious disorder hampering the export of mango variety. It is –

(a) Alphonso (b) Dashehari

(c) Neelum (d) Langra

U.P.P.C.S. (Mains) 2002

Ans. (a)

The spongy tissue, a ripening disorder is often described as soft centre white corky tissue or internal breakdown in Alphonso mangoes. This disorder is peculiar to South India only.

Genetic Engineering and Biotechnology

Notes

- Genetic engineering is the artificial manipulation, modification and recombination of DNA or other nucleic acid molecules in order to modify the genome of an organism or population of organisms using biotechnology.
- This branch of biology is helpful in **genetic library**, **gene therapy** and to produce **transgenic** or **genetically modified organism**.

Genetically Modified Organisms (GMO) :

- This term means that the genome or genetic makeup of the organism has been modified or altered using genetic engineering techniques.
- The process to produce a GMO or transgenic organism involves taking genes from one organism and putting them into another in ways that would not occur naturally.
- Bacteria are often used to grow and reproduce desired genes. Bacteria are great for this because they grow and reproduce quickly and can be controlled quite well in the laboratory. An example of this is insulin production. **Insulin** is a hormone that's normally produced in the human pancreas. Those who are diabetic cannot make it, and another source is needed to treat these patients. The gene for insulin is isolated, inserted into bacteria, produced at a rapid rate and then extracted to be given to diabetics.

- Another way that genes can be transferred from one organism to another is through **embryonic stem cell transfer**. Stem cells are very special because they can differentiate into many other types of cells from an organism, are isolated and grown in the lab. The desirable gene is inserted into the cells and cells are then resorted back into the organism which is then inserted into a surrogate mother where it will develop.
- Microinjection is yet another way that genes can be transferred from one organism to another. Here, the desired gene is injected into the nucleus of an egg which is where we find an organism's DNA. The egg is then implanted into a surrogate mother to develop an organism.
- In genetic engineering, plasmid is used.

Plasmid :

- A plasmid is a small DNA molecule within a cell that is physically separated from chromosomal DNA and can replicate independently.
- They are most commonly found as small circular, double-stranded DNA molecules in bacteria; however, plasmids are sometimes present in archaea and eukaryotic organisms.
- Often the genes carried in plasmids provide genetic advantages, such as antibiotic resistance.
- Scientists have taken advantage of plasmids to use them as tools to clone, transfer and manipulate genes.
- Plasmids that are used experimentally for these purposes are called vectors.
- DNA fragments or genes are inserted into a plasmid vector, creating a so-called recombinant plasmid. This plasmid can be introduced into a bacterium by way of the process called transformation. Then, because bacteria divide rapidly, they can be used as factories to copy DNA fragments in large quantities.
- **pBR 322** is a plasmid and was one of the first widely used E-coli cloning vectors. Created in 1977 in the laboratory of Herbert Boyer at the University of California, SanFrancisco. It was named after the post doctoral researchers who constructed it. The 'p' stands for 'plasmid' and 'BR' for **'Bolivar'** and **'Rodriguez'**.

Restriction enzyme :

- A **restriction enzyme** or restriction endonuclease is an enzyme that cleaves DNA into fragments at or near specific recognition sites within molecule known as restriction sites. Restriction enzymes are one class of the broader endonuclease group of enzymes.
- **Werner Arber :** A Swiss microbiologist and geneticist along with American researchers Hamilton O. Smith and Daniel Nathans, shared the 1978 Nobel Prize in Physiology (Medicine) for the discovery of restriction nucleases in Haemophilus influenzae.

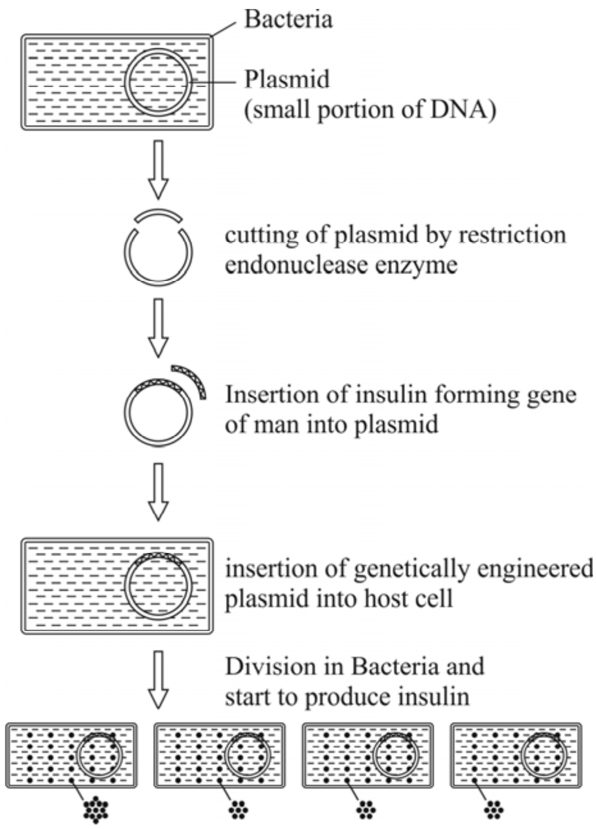

Fig.: Application of plasmid for production of insulin in genetic engineering

DNA ligase :

- DNA ligase is a specific type of enzyme that facilitates the joining of DNA strands together by catalyzing the formation of a phosphodiester bond.
- These two enzymes are used for the production of recombinant DNA in genetic engineering.

Recombinant DNA Technology :

- This technology is related to, joining together of DNA molecules from two or more different sources that are inserted into a host organism to produce new genetic combinations that are of value to science, medicine, agriculture and industry.
- Basic steps involved in recombinant DNA technology are as follows :

  Step I – Identification and isolation of gene of interest.

  Step II – Joining of this gene into a suitable vector (construction of recombinant DNA).

  Step III – Introduction of this vector into a suitable organism.

  Step IV – Selection of transformed recombinant cells with a gene of interest.

  Step V – Multiplication or expression of the gene of interest.

- It is noteworthy that recombinant DNA technology is used to transfer the genes of bacteria or other microorganisms to higher organisms.
- Deoxyribonucleic acid (DNA), is the hereditary material in humans and almost all other organisms. Nearly every cell in a person's body has the same DNA.
- Most DNA is located in the cell nucleus (nuclear DNA) but a small amount of DNA can also be found in the mitochondria and chloroplast (in plant cell).
- The information in DNA is stored as a code made up of four chemical bases : **Adenine** (A), **Guanine** (G), **Cytosine** (C) and **Thymine** (T).
- Human DNA consists of about 3 billion bases and more than 99% of those bases are the same in all people.
- The order of sequence of these bases determines the information available for building and maintaining the organism.
- DNA bases pairup with each other, A with T by two hydrogen bonds and C with G by three hydrogen bonds, to form units called **base pairs**.
- Each base is also attached to a sugar molecule and a phosphate molecule forming a **nucleotide**. Nucleotides are arranged in two long strands, that form a spiral called a **double helix**, which structure is somewhat like a ladder, with the base pairs forming the ladder's rung and the sugar and phosphate molecules forming the vertical side pieces of the ladder.
- For the discovery of the model of DNA as mentioned above, **Watson** and **Crick** shared the Nobel Prize in 1962, along with **Maurice Wilkins**.
- It is noteworthy that **Arthur Kornberg** synthesized DNA in vitro.

CRISPR-Cas9 Gene Editing

- **CRISPR-Cas9** technology is a simple yet powerful tool for editing genomes. It allows researchers to easily alter DNA sequences and modify gene function.
- Its many potential applications include correcting genetic defects, treating and preventing the spread of diseases and improving crops. However, its promise also raises ethical concerns.
- In CRISPR-Cas9 the **'CRISPR'** stands for 'clusters of regularly interspaced short palindromic repeats', which are specialized stretches of DNA. While the protein **'Cas9'** is an enzyme that acts like a pair of **molecular scissors**, capable of cutting strands of DNA.
- This technology was adapted from the natural defence mechanisms of bacteria and archaea.
- By delivering the Cas9 nuclease complexed with a synthetic guide RNA (gRNA) into a cell, the cell's genome can be cut at a desired location, allowing existing genes to be removed and/or new ones added.

Zygote :

- A zygote is a eukaryotic cell (a cell with a nucleus) formed by a fertilization event between two gametes i.e. sperm and oocyte.
- The zygote's genome is a combination of the DNA in each gamete and contains all of the genetic information necessary to form a new individual.
- The zygote divides by mitosis to produce identical offspring.

DNA Fingerprinting :

- The human DNA has four types of nitrogen base with different sequences. But the nitrogen base sequence of all the cells of an individual is similar, which is identical to his family members. On this ground of the sequences of nitrogen bases, the procedure to identify any individual is known as DNA Fingerprinting.
- **Sir Alec John Jeffreys** is a British geneticist who developed techniques for genetic fingerprinting and DNA profiling which are now used worldwide in Forensic Science to assist police detective work and to resolve paternity and immigration disputes. It is also used in judicial cases and for the preservation of threatened livings.
- In the identification of individuals, hair, blood, semen or other biological samples are used.
- **Dr. Lalji Singh** was an Indian scientist who worked in the field of DNA Fingerprinting technology in India. He is the founder of this technology in India and he is called as the **'Father of Indian DNA Fingerprinting'.**
- Colin Pitchfork is a British convicted murderer and rapist. He was the first person convicted of rape and murder based on DNA fingerprinting evidence and the first to be caught as a result of mass DNA screening.
- With help of DNA profiling it may be possible to know the cause and cure of different genetic disorders such as - Alzheimer's (a type of dementia that causes problems with memory, thinking and behaviour), Cystic Fibrosis (a genetic disorder that affects mostly the lungs, but also the pancreas, kidneys and intestine) and Myotonic Dystrophy (A long term genetic disorder that affects muscle functions. Symptoms include gradually worsening muscles loss and weakness. Muscles often contract but unable to relax) etc.

Biometric Identification :

- Biometric identification is any means by which a person can be uniquely identified by evaluating one or more distinguishing biological traits.

- The uses of biometric across the globe is time and attendance in workforce management, airport security, law enforcement, access control and single sign-on (SSO) and in banking-transaction authentication, etc.
- Biometric identification systems can be grouped on the main physical characteristics that lends itself to biometric identification. Examples include :
- Fingerprint identification
- Hand geometry
- Palm vein authentication
- Retina scan
- Iris scan
- Face recognition
- Signature
- Voice analysis
- DNA matching
- Ear size

Biotechnology :

- Biotechnology is the broad area of biology involving living systems and organisms to develop or make products, or "any technological application that uses biological systems, living organisms or derivatives thereof, to make or modify products or processes for specific use".
- Depending on the tools and applications, it often overlaps with the (related) fields of molecular biology, bio-engineering, biomedical engineering, biomanufacturing, molecular engineering, etc.
- The earliest biotechnologists were farmers who developed improved species of plants and animals by cross-pollination or cross-breeding.

Cloning :

- It is a type of genetic engineering that uses cells from one organism to create a second living organism that is genetically identical to the first.
- In nature, many organisms produce clones through asexual reproduction.
- Cloning in biotechnology refers to the process of creating clones of organisms or copies of cells or DNA fragments (molecular cloning).
- The term 'clon' first coined by **Herbert J. Webber** in 1903. It is derived from the Ancient Greek word 'Klon' referring to the process whereby a new plant can be created from a twig. **J.B.S. Haldane** introduced the terms 'clone' and 'cloning' in 1963, modifying the earlier 'clon'.
- **Dolly,** the sheep, a Finn-Dorset ewe, was the first mammal to have been successfully cloned from an adult somatic cell. She was cloned at the Roslin Institute in Scotland and lived there from her birth in 1996 until her death in 2003 when she was six years old. It was cloned by **Keith Campbell**, **Ian Wilmut** and colleagues.
- Dolly was formed by taking a cell from the udder of her 6-year old biological mother. Dolly's embryo was created by taking the cell and inserting it into a sheep ovum. It took 434 attempts before an embryo was successful.
- This embryo later on transferred into the uterus of surrogate sheep, and she gave the birth of Dolly which was identical to her mother (whose cell nucleus was taken).

NDRI's (National Dairy Research Institute, Karnal, Haryana) some cloning milestones

- Feb. 6, 2009 : First male cloned buffalo calf 'Samrupa' born; survives only six days.
- June 6, 2009 : Female cloned buffalo calf 'Garima' takes birth, survives for more than two years, dies on August 18, 2011.
- August 22, 2010 : Female cloned buffalo calf 'Garima-II' born from an embryonic stem cell.
- August 26, 2010 : Male cloned buffalo calf 'Shrestha' born from a somatic cell of an elite bull, produces good quality semen.
- January 25, 2013 : Garima-II delivers female calf 'Mahima'.
- March 18, 2013 : Male cloned buffalo 'Sawarna' born from the somatic cell of semen.
- September 6, 2013 : Female cloned buffalo 'Purnima' born.
- May 2, 2014 : Female cloned buffalo 'Lalima' produced.
- July 23, 2014 : Male cloned buffalo 'Rajat' produced by normal parturition.
- December 27, 2014 : Garima-II gives birth for the second female calf 'Karishma'.
- India's first cloning of Chhattisgarh's endangered wild buffalo met success with the birth of a healthy female named 'Deepasha' on Dec. 12, 2014 at NDRI, Karnal Haryana. Donor mother Asha lives at Udanti wildlife sanctuary, is lone buffalo of its species.
- March 16, 2023 : NDRI produced India's first cloned indigenous Gir breed female cow calf named 'Ganga'.

- It is notable that the first female cloned camel was developed in Camel Breeding Center at Dubai (U.A.E.). Its name was 'Inzaj'. Inzaj was created using the ovarian cells of an adult camel slaughtered in 2005, which were then cultivated in tissue culture and frozen in liquid nitrogen.

- 'Zhong Zhong' (born 27 Nov. 2017) and 'Hua Hua' (born 5 Dec. 2017) are a pair of identical crab-eating macaques (also referred to as cynomolgus monkeys) that were created through somatic cell nuclear transfer (SCNT), the same cloning technique that produced Dolly, the sheep. They are the first cloned primates produced by this technique.

Lulu and Nana Controversy

- The Lulu and Nana controversy revolves around twin Chinese girls born in October 2018, who have been given the pseudonyms **Lulu** and **Nana**. According to the researcher **He Jiankui**, the twins are the world's **first germline genetically edited babies**.
- Gene-editing **CRISPR-Cas9** technique is allegedly used by He Jainkui in creating these babies. It raises many ethical concerns worldwide.

Stem Cell :

- Stem cells are the master cell which acts as basic building blocks of our body. Just like a seed of a plant that gives rise to branches, leaves and fruits, these stem cells have the potential to develop into specialized cells such as blood cells, muscle cells, brain cells etc. of our body.
- When parts of our body or its functions are damaged or affected by ageing, injury or illness, these stem cells have the ability to both replace affected cells and/or repair the affected parts and restore the normal functions of our body.
- All stem cells can self-renew (make copies of themselves) and differentiate (develop into more specialized cells).
- Stem cells have the ability to treat over 80 blood and bone-related conditions, including cancers of the blood, autoimmue disorders and certain genetics disorders.
- Cord blood (umbilical cord blood) is blood that remains in the placenta and in the attached umbilical cord after childbirth. Cord blood is collected because it contains stem cells which can be used to treat hematopoietic and genetic disorders.

Types of Stem Cells :

1. Embryonic Stem Cells :

- Embryonic stem cells are obtained from the inner cell mass of the blastocyst, a mainly hollow ball of cells that, in the human, forms three to five days after an egg cell is fertilized by a sperm. A human blastocyst is about the size of the dot above this 'i'.
- Embryonic stem cells are pluripotent (they can give rise to every cell type in the fully formed body, but not the placenta and umbilical cord).
- Human embryonic stem cells have been derived primarily from blastocysts created by in vitro fertilization (IVF) for assisted reproduction that were no longer needed.

2. Tissue-Specific Stem Cells (Somatic or Adult Stem Cells) :

- These stem cells are more specialized than embryonic stem cells because these stem cells can generate different cell types for the specific tissue or organ in which they live.
- For example, blood-forming (hematopoietic) stem cells in the bone marrow can give rise to RBCs, WBCs, and blood platelets, but these do not generate liver or lung or brain cells, and stem cells in other tissues and organs do not generate RBCs/WBCs/blood platelets.

3. Induced Pluripotent Stem Cells (IPS) :

- These cells have been engineered in the lab by converting tissue-specific cells, such as skin cells, into cells that behave like embryonic stem cells. IPS cells are critical tools to help scientists learn more about normal development and disease onset and progression, and they are also useful for developing and testing new drugs and therapies.

4. Mesenchymal Stem Cells (MSCs) :

- MSCs refer to cells isolated from stroma, the connective tissue that surrounds other tissues and organs.
- These cells are also known as **'stromal cells'**.
- The first MSCs were discovered in the bone marrow and were shown to be capable of making bone, cartilage and fat cells. Since then, they have been grown from other tissues, such as fat and cord blood.

Amniocentesis :

- Amniocentesis (also referred to as amniotic fluid test) is a medical procedure used in prenatal diagnosis of chromosomal abnormalities and fatal infections, and also for sex determination.
- In this procedure, a small amount of amniotic fluid, which contains fetal tissues, is sampled from the amniotic sac surrounding a developing fetus, and then the fetal DNA is examined for genetic abnormalities.
- The most common reason to have an 'amnio' is to determine whether a fetus has a certain genetic disorder or a chromosomal abnormality, such as Down syndrome.
- This procedure is usually performed between 14-16 weeks of pregnancy, though again, can be done at a later stage of gestation as well.
- This procedure can be used for prenatal sex discernment and hence this procedure has legal restriction in many countries.

Applications of Genetic Engineering :

- The application and importance in daily life, of Gene Engineering can be understood by following heads :

1. **Genomic Library :**

- A genomic library contains all the sequences present in the genome (an organism's complete set of DNA, including all of its gene) from a single organism. Each genome contains all of the information needed to build and maintain the organism. In human, a copy of the entire genome-having more than 3 billion nitrogen base pairs - is contained in all cells of the body. In the construction of genomic libraries, it is feasible to use vectors that could accommodate the large size of inserts.
- The first step in the construction of the genomic library is the isolation of the genomic DNA, and entire DNA is subjected to restriction digestion. The fragmented DNA of suitable size is ligated in the appropriate cloning vectors.
- The recombination vectors are transferred and maintained in organisms such as bacteria, virus or yeast. A target DNA sequence present in particular cell clones are identified, sub-cultured and maintained as cell lives, widely known as gene bank or a clone bank.

Human Genome Project

- The Human Genome Project (HGP) was the international collaborative research program whose goal was the complete mapping and understanding of all the genes of human beings. All our genes together are known as our 'genome'.
- As estimated by scientists, the human genome has approximately 30,000 genes and analysis of majority of genes has been completed.
- Human Genome Project will help in -

(i) Study all the genes in a genome.
(ii) Genes concerned with cancer can be found and sequenced study the transcripts in a particular tissue or organ of the tumour.
(iii) Study of interaction of various genes, proteins and their interaction.

2. **Gene Therapy :**

- The main reason for approximately 3000 diseases in human is the synthesis of fault enzymes under control of faulty genes. Under gene therapy, we can cure such diseases.
- **Gene therapy includes :**

i. Replacing a mutated gene that causes disease with a healthy copy of the gene.
ii. Inactivating or knocking out a mutated gene that is functioning improperly.
iii. Introducing a new gene into the body to help fight a disease.

Pleiotropism

- This is the condition of a gene affecting more than one characteristic of the phenotype.
- Example of pleiotropism is Sickle Cell Anaemia of RBC - In this disease, the red blood cells become rigid and sticky and are shaped like sickles or crescent moon. A recessive gene is responsible for sickle cell anaemia.

3. **High Resistance in Plants :**

- Genetically transformed plants are grown by tissue culture techniques and extra desired genes are inserted along with natural genes by genetic engineering.
- The inserted extra desired genes produce the capacity to resist against salinity of soil, drought and insects, virus and bacteria.
- Through genetic engineering, some varieties have been produced that could directly fix atmospheric nitrogen and thus there is no dependence on fertilizers.
- Bacterium, **Bacillus thuringiensis (Bt)** produced a protein which is toxic to insects. Using the technique of genetic engineering, the gene coding for this toxic protein called Bt gene has been isolated from the bacterium and engineered into tomato and tobacco plants. Such transgenic plants showed resistance to tobacco hornworms and tomato fruitworms.
- In Bt Brinjal the gene of B. thuringiensis (Cry1Ac) has been introduced which produces Bt toxin, which kills the lepidopteron insects such as brinjal fruit-borer and shoot-borer.
- This may be proved a good alternative for crop protection but it is a doubt that the poison effect may be harmful to the health of man and biodiversity, so the Government of India has banned the field trials of Bt Brinjal in 2010.
- By the similar technology transgenic cotton (Bt cotton) and transgenic Maize (Bt Maize) have been produced.

4. **Change in Plant Genotype :**

- Useful and good quality of plants like tomato, tobacco, onion, maize, wheat, barley, rice, pea etc. can be developed by inserting rDNA (Recombinant DNA) & cloned DNA into the genotype of plants.

Flavr Savr :

- Flavr Savr also known as CGN-89564-2, a genetically modified tomato was the first commercially grown genetically engineered food to be granted a licence for human consumption.
- It was produced by the Californian company Calgene, and submitted to the U.S. Food and Drug Administration in 1992.
- Through genetic engineering, Calgene hoped to slow down the ripening process of the tomato and thus prevent it from softening, while still allowing the tomato to retain its natural colour and flavour.

- The Flavr Savr was made more resistant to rotting by adding an antisense gene which interferes with the production of the enzyme polygalacturonase.

Golden Rice :

- Golden rice is a variety of rice **(Oryza sativa)** produced through genetic engineering to biosynthesize **beta-carotene**, a precursor of vitamin A in the edible parts of the rice.
- It is intended to produce a fortified food to be grown and consumed in areas with a shortage of dietary vitamin A.
- Golden rice differs from its parental strain by the addition of two beta-carotene biosynthesis genes. The two beta-carotene biosynthesis genes are : psy (phytoene synthase) from daffodil and crtl (phytoene desaturase) from he soil bacterium Erwinia uredovora.

Question Bank

1. 'Microsatellite DNA' is used in the case of which one of the following?

(a) Studying the evolutionary relationships among various species of fauna
(b) Stimulating 'stem cells' to transform into diverse functional tissues
(c) Promoting clonal propagation of horticultural plants
(d) Assessing the efficacy of drugs by conducting series of drug trials in a population

I.A.S. (Pre) 2023

Ans. (a)

Microsatellite, as related to genomics, refers to a short segment of DNA, usually one to six or more base pairs in length, that is repeated multiple times in succession at a particular genomic location. These DNA sequences are typically non-coding. The number of repeated segments within a microsatellite sequence often varies among people, which makes them useful as polymorphic markers for studying inheritance patterns in families or for studying the evolutionary relationships among various species of fauna. They are widely used for DNA profiling in cancer diagnosis, in kinship analysis (especially paternity testing) and in forensic identification. They are also used in genetic linkage analysis to locate a gene or a mutation responsible for a given trait or disease. Microsatellites are also used in population genetics to measure levels of relatedness between subspecies, groups and individuals.

2. With reference to the recent developments in science, which one of the following statements is not correct?

(a) Functional chromosomes can be created by joining segments of DNA taken from cells of different species.
(b) Pieces of artificial functional DNA can be created in laboratories.
(c) A piece of DNA taken out from an animal cell can be made to replicate outside a living cell in a laboratory.
(d) Cells taken out from plants and animals can be made to undergo cell division in laboratory petri dishes.

I.A.S. (Pre) 2019

Ans. (a)

Among the given statements, statement of option (a) is not correct, because functional chromosomes cannot be created by joining segments of DNA taken from cells of different species (through recombinant DNA technique only functional DNA can be created). Other three statements are correct. Statement of option (b) in context of artificial DNA synthesis, statement of option (c) in context of cloning and statement of option (d) in context of tissue culture are correct.

3. Which of the following is the first human hormone produced by recombinant DNA technology?

(a) Estrogen (b) Testosterone
(c) Insulin (d) Thyroxine

Uttarakhand P.C.S. (Pre) 2024

Ans. (c)

The first human hormone produced using recombinant DNA technology was insulin. In 1982 US Food and Drug Administration approved Humulin, Eli Lily's recombinant insulin made from Genentech's specially modified bacteria. It was the first drug produced through recombinant DNA technology and among the first genetically engineered products to be available to consumers.

4. With reference to recent developments regarding 'Recombinant Vector Vaccines', consider the following statements :

1. Genetic engineering is applied in the development of these vaccines.
2. Bacteria and viruses are used as vectors.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2021

Ans. (c)

Live recombinant vector vaccines are made of a live attenuated viral or bacterial strain used as a vector to carry the gene or genes encoding the desired vaccine antigens. In these vaccines live viral or bacterial vector is genetically engineered to express a variety of exogenous antigens in the cytoplasm of targeted T cells. Live recombinant vector vaccines have a number of attractive features, including the ability to stimulate both humoral and cell-mediated immunity. Recombinant vector vaccines (platform-based vaccines) act like a natural infection, so they're especially good at teaching the immune system how to fight germs.

5. **Recombinant DNA Technology (Genetic Engineering) allows genes to be transferred –**
   1. **Across different species of plants.**
   2. **From animals to plants.**
   3. **From microorganisms to higher organisms.**

   Select the correct answer using the codes given below:

   (a) Only 1 (b) 2 and 3
   (c) 1 and 3 (d) All of these

I.A.S. (Pre) 2013

Ans. (d)

Recombinant DNA Technology is used to cut a known DNA sequence from one organism and introduce it into another organism thereby altering the genotype of the recipient. Foreign DNA sequences can be introduced into bacteria, yeast, viruses, plant and animal cells. Thus, genetic engineering (or recombinant DNA technology) allows selected individual gene sequences to be transferred from an organism into other and also between non-related species.

6. **Steps of Recombinant DNA technology are given below:**

   A. Identification and isolation of the genetic material.
   B. Fragmentation of DNA.
   C. Obtaining the foreign gene product.
   D. Downstream processing.
   E. Ligation of DNA fragmentation into the vector.
   F. Isolation of desired DNA fragments.
   G . Amplification of gene of interest.
   H. Transfer of Recombinant DNA into the host cell/ organism.

   The correct sequence of steps is :

   (a) C → A → B → D → E → F → G → H
   (b) A → D → C → B → E → G → F → H
   (c) A → B → F → G →E → H → C → D
   (d) H → F → G → E → A → D → B → C

R.A.S./R.T.S. (Pre) (Re. Exam) 2013

Ans. (c)

Steps in producing recombination DNA : → Identification and isolation of genetic material → Fragmenting the obtained DNA → Isolation of desired DNA fragments → Amplification of gene of interest → Ligation of DNA fragmentation into the vector → Transfer of recombinant DNA into the host cell/organism → Obtaining the foreign gene product → Downstream processing.

7. **Assertion (A) : Scientists can cut apart and paste together DNA molecules at will, regardless of the source of the molecules.**

   Reason (R) : DNA fragments can be manipulated using restriction endonucleases and DNA ligases.

   Code :

   (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
   (b) Both (A) and (R) are true but (R) is not a correct explanation of (A).
   (c) (A) is true but (R) is false.
   (d) (A) is false but (R) is true.

I.A.S. (Pre) 2001

Ans. (a)

Restriction endonucleases are enzymes that cut a DNA molecule at a particular place. By dividing DNA in small pieces with the genetic engineering (through restriction endonucleases), these pieces can be rejoined by DNA ligase enzyme. The Nobel Prize in Physiology or Medicine, 1978 was awarded jointly to Werner Arber, Daniel Nathans and Hamilton O. Smith for the discovery of restriction endonucleases and their application to problems of molecular genetics.

8. **Which of the following pairs is correctly matched ?**

(a) DNA	–	Molecular Scissors
(b) Ligases	–	Molecular Scissors
(c) Ligases	–	Molecular Stitchers
(d) Restriction Endonucleases	–	Molecular Stitchers

R.A.S./R.T.S. (Pre) (Re. Exam) 2013

Ans. (c)

Ligases, work as molecular stitchers are the class of enzymes that catalyze the formation of covalent bonds and are important in the synthesis of biological molecules such as DNA etc. Ligase also activates the process of repair in the break up biological molecules by catalysing the formation of bonds (covalent bonds) between the broken up fragments. Therefore ligases can be given the name of molecular stitchers.

9. **What is Cas9 protein that is often mentioned in news?**

   (a) A molecular scissors used in targeted gene editing
   (b) A biosensor used in the accurate detection of pathogens in patients
   (c) A gene that makes plants pest-resistant
   (d) A herbicidal substance synthesized in genetically modified crops

I.A.S. (Pre) 2019

Ans. (a)

Cas9 protein (CRISPR associated protein 9) is actually a molecular scissors used in targeted gene editing. Its main function is to cut DNA and therefore it can alter a cell's genome. In CRISPR-Cas9 gene editing technique, the 'CRISPR' stands for 'clustered regularly interspaced short palindromic repeats', which are specialized stretches of DNA, while the 'Cas9' is an enzyme that acts like a pair of molecular scissors capable of cutting strands of DNA. This technology was adapted from the natural defence mechanisms of bacteria and archaea. By delivering the Cas9 molecule complexed with a synthetic guide RNA (gRNA) into a cell, the cell's genome can be cut at a desired location, allowing existing genes to be removed and/or new ones added.

10. 'RNA interference (RNAi)' technology has gained popularity in the last few years. Why?

1. It is used in developing gene silencing therapies.

2. It can be used in developing therapies for the treatment of cancer.

3. It can be used to develop hormone replacement therapies.

4. It can be used to produce crop plants that are resistant to viral pathogens.

Select the correct answer using the code given below:

(a) 1, 2 and 4 (b) 2 and 3 only

(c) 1 and 3 only (d) 1 and 4 only

I.A.S. (Pre) 2019

Ans. (a)

RNA interference (RNAi) is a biological process in which RNA molecules inhibit gene expression or translation, by neutralizing targeted mRNA (messenger RNA) molecules. Hence, it is used in developing gene silencing therapies. RNAi is being explored as a form of treatment for a variety of diseases, including HIV/AIDS, hepatitis, Huntington disease and cancer. Many studies have demonstrated that RNAi can provide a more specific approach to inhibit tumor growth by targeting oncogenes (cancer-related genes). Using RNAi technology, scientists have made major breakthroughs in introducing resistance to many viruses in various crops. Thus, RNAi technology can be used to produce crop plants that are resistant to viral pathogens. Thus statements 1, 2 and 4 are correct, while statement 3 is not correct because in development of hormone replacement therapies there is no role of RNA interference.

11. What is the expanded form of the term 'mRNA' that has been widely discussed since the beginning of the pandemic?

(a) Messenger Ribonucleic Acid

(b) Mutant Ribonucleic Acid

(c) Modified Ribonucleic Acid

(d) Mnemonic Ribonucleic Acid

69th B.P.S.C. (Pre) 2023

Ans. (a)

'mRNA' stands for messenger ribonucleic acid. They are single-stranded molecules that carry genetic code from DNA in a cell's nucleus to ribosomes, which make proteins in the cells. These molecules are called messenger RNA because they carry instructions for producing proteins from one part of the cell to another. Researchers have been studying and working with mRNA vaccines for decades and this technology was used to make some of the COVID-19 vaccines, mRNA vaccines make proteins in order to trigger an immune response. mRNA vaccines have several benefits compared to other types of vaccines, including shorter manufacturing times and, because they do not contain a live virus, no risk of causing disease in the person getting vaccinated.

12. In the context of recent advances in human reproductive technology, 'Pronuclear Transfer' is used for :

(a) fertilization of egg in vitro by the donor sperm

(b) genetic modification of sperm producing cells

(c) development of stem cells into functional embryos

(d) prevention of mitochondrial diseases in offspring

I.A.S. (Pre) 2020

Ans. (d)

'Pronuclear Transfer' is a human reproductive technology, which is used for creating a 'three-parent baby' for the prevention of mitochondrial diseases in offspring. The technique involves intervening in the fertilization process to remove faulty mitochondria which can cause fatal heart problems, liver failure, brain disorders, blindness, muscular dystrophy etc. The treatment is known as 'three-parent' IVF because the babies born from genetically modified embryos, would have DNA from a mother, a father and from a woman donor.

13. In the context of hereditary diseases, consider the following statements :

1. Passing on mitochondrial diseases from parent to child can be prevented by mitochondrial replacement therapy either before or after in vitro fertilization of egg.

2. A child inherits mitochondrial diseases entirely from mother and not from father.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2021

Ans. (a)

Mitochondrial Replacement Therapy (MRT), sometimes called mitochondrial donation, is the replacement of mitochondria in one or more cells to prevent or ameliorate disease. MRT originated as a special form of in vitro fertilisation in which some or all of the future baby's mitochondrial DNA (mt DNA) comes from a third party. This technique is used in cases when mothers carry genes for mitochondrial diseases. The therapy is approved for use in the United Kingdom. At present, there are three MRT techniques in use : maternal spindle transfer (MST); pronuclear transfer (PNT); and the newest technique, polar body transfer (PBT). Mitochondrial donation can be performed either prior to or shortly after fertilisation. In both cases, this is before the fertilized egg becomes an embryo. Hence, statement 1 is correct.
Mitochondrial diseases may be caused by mutations (acquired or inherited) in mitochondrial DNA (mtDNA), or in nuclear genes that code for mitochondrial components. Nuclear DNA has two copies per cell (except for sperm and egg cells), one copy being inherited from the father and the other from the mother. Mitochondrial DNA, however, is inherited from the mother only (with some exceptions). Typically, humans inherit mitocondria and mitochondrial DNA from their mothers only. But researchers have identified some individuals who inherited mtDNA from both parents (as per a research paper published in November 2018). Hence, statement 2 is incorrect.
Note: UPSC has given answer (c) for this question in its official answer key.

14. Assertion (A) : Dolly was the first cloned mammal.
Reason (R) : Dolly was produced by in vitro fertilization.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true but (R) is not a correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

I.A.S. (Pre) 1999

Ans. (c)

Dolly was born in Roslin Institute of Scotland on July 5, 1996 by the efforts of Ian Wilmut, Keith Campbell and their colleagues. She was the first mammal in the world, cloned from adult somatic cell successfully. Dolly was developed by conventional cloning technique, not by in vitro fertilization. In vitro fertilization (IVF) is accepted widely for effective treatment of infertility.

15. Which of the facts about Dolly (sheep), the first mammal cloned from an adult somatic cell is not correct?

(a) Dolly died in the year 2003
(b) Dolly died due to lung disease
(c) Dolly was born in the year 1998
(d) Dolly was born in Scotland.

R.A.S./R.T.S. (Pre) 2016

Ans. (c)

Dolly was a female domestic sheep, and the first mammal cloned from an adult somatic cell. She was cloned at University of Edinburgh, Scotland. She was born on 5 July, 1996 and died on 14 February, 2003 from a progressive lung disease 5 months before her seventh birthday.

16. Which of the following is the first living cloned genetic engineered organism by human?

(a) Dolly (b) Herman bull
(c) Bony (d) Super bug

U.P. Lower Sub. (Pre) 2002

Ans. (a)

See the explanation of above question.

17. Which among the following was the first successfully cloned animal?

(a) Dog (b) Rabbit
(c) Gibbon (d) Sheep

U.P. R.O./A.R.O. (Mains) 2021

Ans. (d)

See the explanation of above question.

18. Which one was the first successful cloned animal?

(a) Sheep (b) Gibbon
(c) Rabbit (d) None of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (a)

See the explanation of above question.

19. Scientists of N.D.R.I., Karnal (Haryana) developed the second clone of which on the following animals ?

(a) Sheep (b) Buffalo
(c) Cow (d) Goat

M.P.P.C.S. (Pre) 2012

Ans. (b)

The world's first cloned buffalo calf 'Samrupa' through the 'Hand guided Cloning Technique' was born on 6 February, 2009 at NDRI, Karnal and subsequently, the second cloned calf 'Garima' was born on 6 June, 2009 with the birth weight of 43 kg.

20. Garima II is the name of a –

(a) Cloned buffalo (b) Cloned cow
(c) Cloned sheep (d) Bt tomato

U.P.P.C.S. (Mains) 2009

Ans. (a)

The team of scientists of NDRI situated in Karnal got success to develop a clone of buffalo, named as Garima II on 22 Aug, 2010. Clone buffalo Garima II gave birth a healthy female calf Mahima in NDRI on 25 January, 2013.

21. 'Ganga', India's first cloned cow born in March, 2023 belongs to which breed?

(a) Sahiwal (b) Gir
(c) Tharparkar (d) Nagauri

Rajasthan P.C.S. (Pre) 2024

Ans. (b)

The Karnal-based National Dairy Research Institute (NDRI) has cloned, for the first ever in India, a female calf of the indigenous cow breed Gir that was capable of producing over 15 litres of milk per day. India's first cloned Gir female calf is named 'Ganga'. She was born on 16 March, 2023 weighing 32 kg.

22. Which country has produced the first transgenic glowing pigs that are all green from inside out?

(a) Korea (b) Japan
(c) Singapore (d) Taiwan

U.P.P.S.C.(GIC) 2010

Ans. (d)

The research group of National Taiwan University produced first transgenic glowing pig in 2006. Three male transgenic pigs were born by introducing glowing green protein in the embryo of pig.

23. Injaz, is the name of the world's first cloned :

(a) Camel (b) Goat
(c) Pig (d) Sheep

U.P.P.C.S. (Mains) 2008

Ans. (a)

World's first female cloned camel named as Injaz developed in Camel Reproduction Centre, Dubai, United Arab Emirates.

24. Which of the following statements about a clone of an organism is correct?

(a) A clone has characteristics of both its parents
(b) A clone is produced asexually
(c) Identical twins are clones of an organism
(d) Two clones of an organism may not be identical

R.A.S./R.T.S.(Pre) 2003

Ans. (b)

Any organism whose genetic information is identical to that of a parent organism is known as a clone. They have produced asexually and are identical to parent in all Any organischaracters. The process to obtain clone is called as cloning. Identical twins are not clones and two clones of an organism are identifical.

25. In the case of a test-tube baby –

(a) Fertilization takes place inside the test-tube
(b) Development of the baby takes place inside the test-tube
(c) Fertilization takes place outside the mother's body
(d) Unfertilised egg develops inside the test-tube.

I.A.S. (Pre) 1994

Ans. (c)

Females, who are not able to conceive a natural pregnancy, their eggs are artificially fertilized. The child born by IVF technique is known as 'Test-tube baby'. Sperm-egg fertilization, which happens naturally in female fallopian tubes, takes place in Petri dishes artificially outside of mother's body. After these fertilized embryos are implanted in the uterus of a female for getting new offspring.

26. Consider the following effects of genetic engineering:

1. Disease resistance 2. Growth promotion
3. Animal cloning 4. Human cloning

Of the above, that have been tried with a certain amount of success include :

(a) 1, 3 and 4 (b) 2, 3 and 4
(c) 1, 2 and 4 (d) 1, 2 and 3

U.P.P.C.S. (Pre) 2003
U.P.P.C.S.(Pre) 2001

Ans. (d)

Through genetic engineering, disease resistance, growth enhancement and animal cloning have been achieved successfully but human cloning has not got proper fruitful results because it is controversial and banned.

27. Match List-I with List-II and select the correct answer using the codes given below :

List-I (Achievement in Genetics)

A. Discovery of transduction and conjugation in bacteria
B. Establishing the sex-linked inheritance
C. Isolation of DNA polymerase from E. coli
D. Establishing the complete genetic code

List-II (Scientists)

1. Khorana 2. Korenberg
3. Lederberg 4. Morgan
5. Ochoa

Code :

	A	B	C	D
(a)	4	3	2	1
(b)	3	4	1	5
(c)	4	3	1	5
(d)	3	4	2	1

I.A.S. (Pre) 2001

Ans. (d)

Bacterial conjugation is the transfer of genetic material between bacterial cells by direct cell-to-cell contact or by a bridge-like connection between two cells, discovered in 1946 by Joshua Lederberg and Edward Tatum. Bacterial transduction was discovered in Salmonella by Norton Zinder and Joshua Lederberg in 1952. Morgan was the first to establish the chromosome theory of inheritance.He used fruit flies with eye colour mutations to demonstrate sex-linked inheritance patterns. Discovered by Arthur Kornberg in 1956, it was the first known DNA polymerase (and, indeed, the first known of any kind of polymerase). It was initially characterized in E. coli and is ubiquitous in prokaryotes. Hargobind Khorana was an Indian-American biochemist who shared the 1968 Nobel Prize in Physiology or Medicine with Marshall W. Nirenberg and Robert W. Holley for research that helped to show how the order of nucleotides in nucleic acids, which carry the genetic code of the cell, control the cell's synthesis of proteins.

28. At present, scientists can determine the arrangement or relative positions of genes or DNA sequences on a chromosome. How does this knowledge benefit us?

1. It is possible to know the pedigree of livestock .

2. It is possible to understand the causes of all human diseases.

3. It is possible to develop disease-resistant animal breeds.

Which of the statements given above is/are correct ?

(a) 1 and 2 (b) 2 only

(c) 1 and 3 (d) 1, 2, and 3

I.A.S. (Pre) 2011

Ans. (c)

Treatment of different genetic diseases e.g. Alzheimer, cystic fibrosis, myotonic dystrophy etc. can be possible with the help of D.N.A. sequencing but this cannot help to understand the causes of all human diseases. Thus, statement 2 is not correct. Other two statements are correct.

29. The sequencing of the entire genes of an organism was done in 1996. That organism was :

(a) Albinistic mouse (b) Yeast

(c) Human being (d) Plasmodium vivax

I.A.S. (Pre) 1997

Ans. (b)

The genome of the budding yeast Saccharomyces cerevisiae was first completely sequenced from a eukaryote. It was released in 1996 as the work of a worldwide effort of hundreds of researchers.

30. World level program 'Human Genome Project' is related with –

(a) Establishment of Superman society

(b) Identification of colour distinct breeds

(c) Genetic improvements of human breeds

(d) Identification and mapping of human genes and its sequence

U.P.P.C.S. (Pre) 2001

Ans. (d)

World level programme 'Human Genome Project' is related to identification and mapping of human genes and its sequences.

31. The cells which have the capacity to divide and differentiate into any type of cells in the body are the focus of research of several serious diseases are –

(a) Bud cells (b) Red cells

(c) Mesangial cells (d) Stem cells

U.P.P.C.S. (Pre) 2006

Ans. (d)

Stem cells are unique as they have the potential to develop into many different cell types in the body, including brain cells, but they also retain the ability to produce more stem cells, a process termed as self–renewal. There are multiple types of stem cells such as embryonic stem cells, induced pluripotent stem cells and adult or somatic stem cells.

32. With reference to 'stem cells', frequently in the news, which of the following statement(s) is/are correct?

1. Stem cells can be derived from mammals only.

2. Stem cells can be used for screening new drugs.

3. Stem cells can be used for medical therapies.

Select the correct answer using the codes given below:

(a) 1 and 2 (b) 2 and 3

(c) Only 3 (d) 1, 2 and 3

I.A.S. (Pre) 2012

Ans. (b)

Stem cells are undifferentiated biological cells that can differentiate into specialized cells and can divide (through mitosis) to produce more stem cells. They are found in multi–cellular organisms. In mammals, there are two broad types of stem cells : embryonic stem cells, which are isolated from the inner cell mass of blastocysts and adult stem cells, which are found in various tissues. Stem cells can be derived from mammals and also from other multi-cellular organisms. Stem cells are frequently used in various medical therapies (e.g. bone marrow transplantation) and can be used for screening new drugs.

33. With reference to the latest developments in stem cell research, consider the following statements :

1. The only source of human stem cells are the embryos at the blastocyst stage.

2. The stem cells can be derived without causing destruction to blastocysts.

3. The stem cells can regenerate themselves in vitro virtually forever.

4. Indian research centres also created a few cell lines which can be developed into many types of tissues.

Which of these statements are correct ?

(a) 1, 2, 3 and 4 (b) 1, 2 and 3

(c) 1, 3 and 4 (d) 3 and 4

I.A.S. (Pre) 2002

Ans. (d)

Based on the cell type/tissue of origin, stem cells are classified as 'Somatic Stem Cells' (SSCs) and 'Embryonic Stem Cells' (ESCs). SSCs could be obtained from different sources, for example the fetus, umbilical cord, placenta, infant, child or adult; and from different organs/tissues while ESCs are derived from pre-implantation embryos (blastocysts). The derivation of human embryonic stem cells currently requires the destruction of ex-utero embryos. The stem cells can regenerate themselves in vitro virtually forever. Indian research centres also created a few cell lines which can be developed into many types of tissues. Thus, only statements 3 and 4 are correct.

34. Which one of the following statements is not true with regard to the transplantation of stem cells in animals?

(a) They multiply themselves throughout the life span of an animal

(b) They repair the damaged tissues of the organs

(c) They have the capacity to produce one or more types of specialized cells

(d) They are found only in the embryo

R.A.S./R.T.S.(Pre) 2007

Ans. (d)

See the explanation of above question.

35. Which one of the following is the bioethically non-controversial source of stem cells as an alternative to the highly controversial embryonic stem cells?

(a) Bone marrow-derived stem cells

(b) Amniotic fluid derived stem cells

(c) Blood of foetus

(d) Blood of babies

U.P.P.C.S. (Pre) 2008

Ans. (a)

Stem cells are undifferentiated biological cells that can differentiate into specialized cells and can divide (through mitosis) to produce more stem cells. They are found in multicellular organisms. There are three known accessible sources of autologous adult stem cells in humans : Bone marrow, which requires extraction by harvesting that is drilling into bone (typically the femur or iliac crest). This is an ultimate alternative to controversial embryonic stem cells. Adipose tissue (lipid cells), which requires extraction by liposuction. Blood, which requires extraction through aphaeresis, wherein blood is drawn from the donor (similar to a blood donation) and passed through a machine that extracts the stem cells and returns other portions of the blood to the donor.

36. Consider the following statements:

1. Genetic changes can be introduced in the cells that produce eggs or sperms of a prospective parent.

2. A person's genome can be edited before birth at the early embryonic stage.

3. Human induced pluripotent stem cells can be injected into the embryo of a pig.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 and 3 only

(c) 2 only (d) 1, 2 and 3

I.A.S. (Pre) 2020

Ans. (d)

By using the gene-editing tool CRISPR-Cas9, genetic changes can be introduced in the cells that produce eggs or sperms of a prospective parent, and a person's genome can be edited before birth at the early embryonic stage. Researchers from the Oregon Health and Science University, USA along with colleagues in California, China and South Korea had demonstrated that by repairing a mutation in human embryos using CRISPR-Cas9. Further, researchers from Spain, China and Japan, etc. had shown that human induced pluripotent stem cells could be injected into the embryo of a pig (or mice, rats). In March, 2019 Japan changes its rules on implanting human cells into animals.

37. What is the application of Somatic Cell Nuclear Transfer Technology?

(a) Production of biolarvicides

(b) Manufacture of biodegradable plastics

(c) Reproductive cloning of animals

(d) Production of organisms free of diseases

I.A.S. (Pre) 2017

Ans. (c)

Somatic cell nuclear transfer (SCNT) can be used in embryonic stem cell research, or in regenerative medicine where it is sometimes referred to as 'therapeutic cloning'. It can also be used as the first step in the process of reproductive cloning. The nucleus of the somatic cell is inserted into the enucleated (after removal of nucleus) egg cell.

38. Hybridoma technology is a new biotechnological approach for commercial production of :

(a) Monoclonal antibodies (b) Interferon
(c) Antibiotics (d) Alcohol

I.A.S. (Pre) 2000

Ans. (a)

Hybridoma technology is a technology of forming hybrid cell lines (called hybridomas) by fusing an antibody–producing B cell with a myeloma (B cell cancer) cell that is selected for its ability to grow in tissue culture and for an absence of antibody chain synthesis. The antibodies produced by the hybridoma are all of a single specificity and are therefore monoclonal antibodies (in contrast to polyclonal antibodies).The production of monoclonal antibodies through hybridoma technology was invented by Cesar Milstein and Georges J. F. Kohler in 1975. They shared the Nobel Prize in 1984 for medicine and physiology with Niels Kaj Jerne, who made other contributions to immunology.

39. The hybridoma technique was developed by whom?

(a) Louis Pasteur
(b) Georges Kohler and Cesar Milstein
(c) Edward Jenner
(d) Burnet

Uttarakhand P.C.S. (Pre) 2024

Ans. (b)

See the explanation of above question.

40. The Plant Field Gene Bank at Banthara will :

(a) Preserve endangered varieties of plants
(b) Check piracy of bio-diversity
(c) Identify economically important wild plants
(d) Look after all the above

U.P.P.C.S.(Pre) 2001, 2003

Ans. (d)

The Plant Field Gene Bank at Banthara will secure the endangered plants, prevent the piracy of biological diversity and also recognise the economically important plants.

41. Consider the following statements :

1. According to the Indian Patents Act, a biological process to create seed can be patented in India.

2. In India, there is no Intellectual Property Appellate Board.

3. Plant varieties are not eligible to be patented in India.

Which of the statements given above is/are correct?

(a) 1 and 3 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2019

Ans. (c)

According to article 3 (j) of The Patents Act, 1970 (of India) "plants and animals in whole or any part thereof other than micro organisms but including seeds, varieties and species and essentially biological processes for production or propagation of plants and animals" are not inventions and not eligible to be patented in India. Hence, statement 1 is not correct while statement 3 is correct. Statement 2 is not correct because 'Intellectual Property Appellate Board', a quasi-judicial body, was constituted in India in September, 1958 to decide the disputes under the Copyright Act, 1957. Another 'Intellectual Property Appellate Board' has been constituted on 15th September, 2003 to hear appeals against the decisions of the Registrar under the Trade Marks Act, 1999 and the Geographical Indications of Goods (Registration and Protection) Act, 1999.

42. Insect-resistant cotton plants have been genetically engineered by inserting a gene from a/an :

(a) Virus (b) Bacterium
(c) Insect (d) Plant

I.A.S. (Pre) 2000

Ans. (b)

Insect-resistant cotton plants have been genetically engineered by inserting a gene from a bacteria (Bacillus thuringiensis). Strains of Bacillus thuringiensis produces different toxins which are harmful to different insects. Such variety of cotton is commonly known as Bt cotton. The American multinational company Monsanto has produced such variety.

43. The American multinational company, Monsanto, has produced an insect-resistant cotton variety that is undergoing field trials in India. A toxin gene from which one of the following bacteria has been transferred to this transgenic cotton ?

(a) Bacillus subtilis
(b) Bacillus thuringiensis
(c) Bacillus amyloliquifaciens
(d) Bacillus globlii

I.A.S. (Pre) 2001

Ans. (b)

See the explanation of above question.

44. The micro-organism which is associated with the production of Bt cotton is a

(a) Fungus (b) Bacterium
(c) Blue-green algae (d) Virus

U.P.P.C.S (Pre) 2010

Ans. (b)

See the explanation of above question.

45. Assertion (A) : Insect resistant transgenic cotton has been produced by inserting Bt gene.

Reason (R) : The Bt gene is derived from a bacterium.

Code :

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true (R) is not a correct explanation of (A).
(c) (A) is true but (R) is true.
(d) (A) is false but (R) is true.

I.A.S. (Pre) 1999

Ans. (a)

See the explanation of above question.

46. Bollgard I and Bollgard II technologies are mentioned in the context of :

(a) Clonal propagation of crop plants
(b) Developing genetically modified crop plants
(c) Production of plant growth substances
(d) Production of biofertilizers

I.A.S. (Pre) 2021

Ans. (b)

Bollgard I Bt cotton (single-gene technology) is India's first biotech crop technology approved for commercialization in India in 2002, followed by Bollgard II – double-gene technology in mid-2006, by the Genetic Engineering Approval Committee (GEAC), the Indian regulatory body for biotech crops.
Bollgard I cotton provides in-built protection for cotton against destructive American Bollworm Heliothis Armigera infestations, and contains an insecticidal protein from a naturally occurring soil microorganism, Bacillus thuringiensis (Bt). Bollgard II technology contains a superior double-gene technology – Cry 1 Ac and Cry 2 Ab which provides protection against bollworms and Spodoptera caterpillar, leading to better boll retention, maximum yield, lower pesticides costs, and protection against insect resistance. Both, Bollgard II and Bollgard I insect-protected cotton are widely planted around the world as an environment-friendly way of controlling bollworms.

47. A genetically engineered form of brinjal, known as the Bt brinjal has been developed. The objective of this is:

(a) To make it pest-resistant
(b) To improve its taste and nutritive qualities
(c) To make it drought-resistant
(d) To make its shelf-life longer

I.A.S. (Pre) 2011

Ans. (a)

Bt brinjal is a transgenic brinjal created by inserting a gene 'Cry 1 Ac' from the soil bacterium 'Bacillus thuringiensis' into brinjal. Bt brinjal has been developed to give resistance against lepidopteran insects like the brinjal shoot-borer Leucinodes orbonalis and fruit-borer Helicoverpa armigera. Bt brinjal has generated much debate in India. The promoters say that Bt brinjal will be beneficial to small farmers because it is insect resistant, increases yields etc. On the other hand, concerns about Bt brinjal relates to its possible adverse impact on human health and bio-safety, livelihoods and biodiversity. In India, Bt brinjal was developed by the Maharashtra Hybrid Seeds Co. (Mahyco). Despite the conduct of field trials from 2002-2006, a moratorium was issued in October, 2009 and a government ban was implemented in February, 2010.

48. Genetically modified, Bt Brinjal is developed for :

(a) Drought Resistance (b) Insect Resistance
(c) Bacterial Resistance (d) Fungal Resistance

M.P. P.C.S. (Pre) 2022

Ans. (b)

See the explanation of above question.

49. Bt brinjal is –

(a) A new variety of Brinjal
(b) A genetically modified Brinjal
(c) A wild variety of Brinjal
(d) None of the above

Uttarakhand Lower Sub. (Pre) 2010

Ans. (b)

See the explanation of above question.

50. What are the reasons for the people's resistance to the introduction of Bt brinjal in India?

1. **Bt brinjal is created by inserting a gene from soil fungus into its genome.**
2. **The seeds of Bt brinjal are terminator seeds and therefore, the farmers have to buy the seeds before every season from the seed companies.**
3. **There is an apprehension that the consumption of Bt brinjal may have adverse impact on health.**

4. **There is some concern that the introduction of Bt brinjal may have adverse effect on the biodiversity.**

Select the correct answer using the codes given below:

(a) Only 1, 2 and 3 (b) Only 2 and 3
(c) Only 3, 4 (d) 1, 2, 3 and 4

I.A.S. (Pre) 2012

Ans. (c)

See the explanation of above question.

51. With reference to the Genetically Modified mustard (GM mustard) developed in India, consider the following statements :

1. **GM mustard has the genes of a soil bacterium that give the plant the property of pest-resistance to a wide variety of pests.**
2. **GM mustard has the genes that allow the plant cross-pollination and hybridization.**
3. **GM mustard has been developed jointly by the IARI and Punjab Agricultural University.**

Which of the statements given above is/are correct?

(a) 1 and 3 only (b) 2 only
(c) 2 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2018

Ans. (b)

GM mustard developed in India, also known as Dhara Mustard Hybrid-11 (DMH-11), is a genetically modified hybrid variety of the mustard species Brassica juncea. It was developed by Prof. Deepak Pental (University of Delhi) with the aim of reducing India's demand for edible oil imports. His team introduced several genes from a soil bacterium, Bacillus amyloliquefaciens, into the mustard that allow the plant cross-pollination and hybridization. Then they cross-pollinated Indian variety with European variety to make new hybrid called DMH-11. The above DMH-11 has been developed by the scientists of 'Center for Genetic Manipulation of Crop Plants' of Delhi University. Hence, only statement 2 is correct.

52. Other than resistance to pests, what are the prospects for which genetically engineered plants have been created?

1. **To enable them to withstand drought.**
2. **To increase the nutritive value of the produce.**
3. **To enable them to grow and do photosynthesis in space ships and space stations.**
4. **To increase their shelf life.**

Select the correct answer using the code given below:

(a) 1 and 2 (b) 3 and 4
(c) 1, 2 and 4 (d) 1, 2, 3 and 4

I.A.S. (Pre) 2012

Ans. (c)

Genetically modified crops (GMCs, GM crops, or biotech crops) are plants used in agriculture, the DNA of which has been modified using genetic engineering techniques. In most cases, the aim is to introduce a new trait to the plant which does not occur naturally in the species. Examples of such traits in food crops include resistance to certain pests, diseases or environmental conditions, reduction of spoilage or resistance to chemical treatments (e.g. resistance to a herbicide), to enable them to withstand drought, to increase the nutritive value of the produce, to increase their shelf life, or improving the nutrient profile of the crop.

53. With reference to agriculture in India, how can the technique of 'genome sequencing', often seen in the news, be used in the immediate future?

1. **Genome sequencing can be used to indentify genetic markers for disease resistance and drought tolerance in various crop plants.**
2. **This technique helps in reducing the time required to develop new varieties of crop plants.**
3. **It can be used to decipher the host-pathogen relationship in crops.**

Select the correct answer using the code given below :

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2017

Ans. (d)

Genome sequencing plays an important role in the agriculture sector. It can be used to identify genetic indicators, which is essential for developing properties such as disease resistance and drought tolerance in crops. Genome sequencing can help in reducing the time required to develop new varieties of crops. Also, it can be used to understand the host-pathogen relationship in crops.

54. Consider the following statements and choose the correct ones using the codes given below :

A . The flavr-savr tomato was the first genetically engineered crop product to be commercialised.

B . Ripe fruits of flavr-savr remain firm for longer duration and can be transported to market after vine-ripening.

C . Ripe fruits of flavr-savr have colour but not the full array of vine ripened tomato flavours.

Code :

(a) B and C (b) A, B and C
(c) A and B (d) A and C

R.A.S./R.T.S. (Pre) 2016

Ans. (c)

The flavr-savr tomato was the first commercially grown genetically engineered food to be granted a license for human consumption. The novel variety was developed by insertion of an additional copy of the PG (Polygalacturonase) encoding gene in the "antisense" orientation, resulting in reduced translation of the endogenous PG messenger RNA (mRNA). The antisense PG gene is essentially reverse copy of part of the native tomato PG gene that suppresses the expression of endogenous PG enzyme prior to the onset of fruit ripening. Flavr-savr tomatoes have improved harvest and processing properties that allow the transgenic tomatoes to remain longer on the vine to develop their natural flavour, maintain firmness for shipping and produce a thicker consistency in processing.

55. Consider the following techniques/ phenomena :

1. Budding and grafting in fruit plants

2. Cytoplasmic male sterility

3. Gene silencing

Which of the above is/are used to create transgenic crops?

(a) 1 only (b) 2 and 3

(c) 1 and 3 (d) None

I.A.S. (Pre) 2014

Ans. (b)

Cytoplasmic male sterility and gene silencing are techniques used to create transgenic crops. However, budding and grafting do not change the genetic structure of plants.

56. Consider the following kinds of organisms :

1. Bacteria 2. Fungi

3. Flowering plants

Some species of which of the above kinds of organisms are employed as biopesticides?

(a) Only 1 (b) 2 and 3

(c) 1 and 3 (d) 1, 2 and 3

I.A.S. (Pre) 2012

Ans. (d)

Biopesticides fall into three major classes : Microbial pesticides consist of a microorganism (e.g., a bacterium, fungus, virus or protozoan) as the active ingredient. Microbial pesticides can control many different kinds of pests, although each separate active ingredient is relatively specific for its target pests. For example, there are fungi that control certain weeds and other fungi that kill specific insects. A soil-dwelling bacterium, Bacillus thuringiensis (Bt) is the most commonly used biological pesticide worldwide. Neem is the best example as biopesticide among flowering plants.

57. Which one of the following is a microbial insecticide?

(a) Bacillus thuringiensis

(b) Bacillus subtilis

(c) Bacillus polymyxa

(d) Bacillus brevis

Chhatisgarh P.C.S. (Pre) 2018

Ans. (a)

Bacillus thuringiensis (Bt) is a Gram-positive, soil-dwelling bacterium which naturally produces a toxin that is fatal to certain herbivorous insects. The toxin produced by B. thuringiensis has been used as an insecticide spray since 1920 and is commonly used in organic farming. Bt is also the source of the genes used to genetically modify a number of food crops so that they produce toxin on their own to deter various insects or pests. The toxin is lethal to several orders of insects, including Lepidoptera (butterflies, moths and skippers); Diptera (flies) and Coleoptera (beetles).

58. Bacillus thuringiensis is used as :

(a) Biofertilizer (b) Biological insecticide

(c) Chemical fertilizer (d) Chemical insecticide

Uttarakhand P.C.S. (Pre) 2012

Ans. (b)

Bacillus thuringiensis (or Bt) is a Gram-positive, soil-dwelling bacterium, commonly used as a biological pesticide.

59. Indian farmers are unhappy over the introduction of Terminator Seed Technology because the seeds produced by this technology are expected to :

(a) show poor germination

(b) form low-yielding plants despite the high quality

(c) give rise to sexually sterile plants

(d) give rise to plants incapable of forming viable seeds

I.A.S. (Pre) 1999

Ans. (d)

Indian farmers are not happy with Terminator Seed Technology because it increases the chances to grow such plants, which are not capable of producing germinable seeds. This technology works after insertion of a tri-gene. The first gene ends the germination ability of embryo. Recombinase enzyme regulates when this gene will active (which produced from the second gene). The third gene controls over recombinase. Terminator technology is opposed by the farmers throughout the developing world because they would no longer be able to save seeds to re-use from one harvest to next.

60. The controversial terminator technology backed by developed nations seeks to provide :

(a) Biotechnologically improved variety of seeds which are ensured to yield sterile seed for next generation
(b) Transgenic seeds which ensure no-terminating passage of good traits generation after generation
(c) Selective termination of crop diseases
(d) Hybrid seeds that can be eaten but not grown

U.P.P.C.S. (Pre) 1999

Ans. (a)

See the explanation of above question.

61. Terminator technology promotes the sale of which of the following that is/are generated by it ?

(a) Transgenic fertile seed
(b) Gene modified plants
(c) Genetically engineered seeds sterile in next generation
(d) All of the above

U.P.P.C.S. (Mains) 2004

Ans. (d)

See the explanation of above question.

62. Which of the following is a transgenic plant ?

(a) Buckwheat (b) Macaroni wheat
(c) Golden rice (d) Triticale

U.P.P.C.S. (Mains) 2012

Ans. (c)

Golden rice is a transgenic plant. It is a variety of rice (Oryza sativa) produced through genetic engineering to biosynthesize beta-carotene, a precursor of vitamin A, in the edible parts of rice. The research was conducted with the goal of producing a fortified food to be grown and consumed in areas with a shortage of dietary vitamin A, a deficiency which is the world's leading cause of blindness among children. A rice enriched with beta-carotene promises to boost the health of poor children around the world.

63. Assertion (A) : 'Golden rice' is a biotechnological achievement to the benefit of consumers as much as the farmers.

Reason (R) : The yellow colour of this rice reflects a high level of beta carotene, a compound that is converted to vitamin A in the body.

Code :

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

U.P.P.C.S. (Pre) 2000

Ans. (a)

See the explanation of above question.

64. Golden rice is –

(a) Wild variety of rice with yellow coloured grains
(b) A variety of rice grown along the yellow river in China
(c) Long stored rice having yellow colour tint
(d) A transgenic rice having gene for carotene

R.A.S./R.T.S. (Pre) 2013

Ans. (d)

See the explanation of above question.

65. The prime utility of 'Golden Rice' in combating vitamin-A deficiency, the world's leading cause of blindness which affects about 250 million children lies in the richness of its kernel in –

(a) Beta carotene (b) Thiamine
(c) Ascorbic acid (d) Calciferol

U.P.P.C.S. (Mains) 2002

Ans. (a)

See the explanation of above question.

66. Golden rice is rich in –

(a) Vitamin A (b) Vitamin B
(c) Vitamin C (d) Vitamin D

U.P.P.C.S. (Spl) (Mains) 2008

Ans. (a)

See the explanation of above question.

67. Golden rice is a richer source of –

(a) Vitamin A (b) Vitamin B_{12}
(c) Vitamin C (d) Vitamin D

U.P.R.O./A.R.O. (Mains) 2014

Ans. (a)

See the explanation of above question.

68. Golden Rice is a rich source of :

(a) Vitamin A (b) Vitamin B
(c) Vitamin K (d) Vitamin C

U.P. P.C.S. (Mains) 2016
U.P.P.C.S (Pre) 2011

Ans. (a)

See the explanation of above question.

69. For which desirable character the transgenic crop 'Golden Rice' is produced ?

(a) Vitamin 'A'
(b) Essential Amino Acids
(c) Insulin
(d) Characteristic Starch

R.A.S./R.T.S.(Pre) 2008

Ans. (a)

See the explanation of above question.

70. The Golden Rice contains β-carotene gene which comes from –

(a) Carrot (b) Daffodil
(c) Beetroot (d) Papaya

Jharkhand P.C.S. (Pre) 2016

Ans. (b)

Golden Rice is the achievement in the field of biotechnology by Prof. Ingo Potrikus and Dr. Peter Beyer. The colour of this rice is golden. Two genes of daffodil and one gene of Erwinia uredovora is inserted in its genome. These three genes produce the enzymes needed to convert naturally occuring compounds into the immature embryoids of rice, that convert Geranylgeranyl-diphosphate (GGDP) into beta-carotene. On reaching in our body beta-corotene converts into vitamin A which is very useful for our eyes.

71. Which of the following pairs is NOT correctly matched?

(a) Renneting – Cheese
(b) Genetic Engineering – Plasmids
(c) Golden rice – Vitamin A
(d) Ozone layer – Troposphere

U.P. P.C.S. (Pre) 2018

Ans. (d)

The ozone layer or ozone shield is a region of Earth's stratosphere, not trophosphere, that absorbs most of the Sun's ultraviolet radiation. Renneting is the action or process of curding milk by the addition of rennet. Plasmid (found in bacteria) is a genetic tool used in gene engineering. Golden Rice is Genetically Modified or transgenic rice with enriched vitamin A.

72. 'Super rice' was developed by –

(a) M.S. Swaminathan (b) G.S. Khush
(c) N.E. Borlog (d) P.K. Gupta

U.P.P.C.S. (Pre) 2007

Ans. (b)

Super rice was developed by Gurdev Singh Khush, a chief breeder at Philippines-based International Rice Research Institute (IRRI). The research for super high-yielding rice started in 1989. Gurdev Singh Khush is an agronomist and geneticist who along with mentor Henry Beachell, received the 1996 World Food Prize for his achievements in enlarging and improving the global supply of rice during a time of exponential population growth.

73. Amniocentesis is a method used to determine the :

(a) Foetal sex
(b) Kind of amino acids
(c) Sequence of amino acids in protein
(d) Type of hormones

U.P.P.C.S (Pre) 2011

Ans. (a)

Amniocentesis (also referred to as amniotic fluid test or AFT) is a medical procedure used in prenatal diagnosis of chromosomal abnormalities and fetal infections. It is also used for sex determination. In amniocentesis, a small amount of amniotic fluid, which contains fetal tissues is sampled from the amniotic sac surrounding a developing fetus and the fetal DNA is examined for genetic abnormalities.

74. In the context of the developments in Bioinformatics, the term 'transcriptome', sometimes seen in the news, refers to :

(a) A range of enzymes used in genome editing
(b) The full range of mRNA molecules expressed by an organism
(c) The description of the mechanism of gene expression
(d) A mechanism of genetic mutations taking place in cells

I.A.S. (Pre) 2016

Ans. (b)

Transcriptome refers to the sum total of all the messenger RNA (mRNA) molecules expressed from the genes of an organism. Unlike the genome, which is roughly fixed for a given line (excluding mutations), the transcriptome can vary with external environmental conditions.

75. Who synthesized the DNA in vitro?

(a) Arthur Kornberg (b) Robert Hooke
(c) Edward Jenner (d) Joseph Lister

56th to 59th B.P.S.C. (Pre) 2015

Ans. (a)

Arthur kornberg synthesized the DNA in vitro. He won the 1959 Nobel prize in Physiology or Medicine for his discovery of the mechanisms in the biological synthesis of DNA.

76. **Consider the following statements :**

1. **Biofilms can form on medical implants within human tissues.**
2. **Biofilms can form on food and food processing surfaces.**
3. **Biofilms can exhibit antibiotic resistance.**

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2022

Ans. (d)

Biofilm is a complex structure of microbiome having different bacterial colonies or single type of cells in a group, adhere to the surface. It is a microbial derived sessile community characterized by cells that are irreversibly attached to a substratum or interface to each other, embedded in a matrix of extracellular polymeric substances that they have produced. A significant proportion of medical implants become the focus of a device-related infection, difficult to eradicate because bacteria that cause these infections live in well-developed biofilms. Biofilms are also a constant concern in food processing environments as they can form on food and food processing surfaces. Bacterial biofilms are serious global health concern due to their abilities to tolerate antibiotics, host defence systems and other external stresses. Biofilm matrix gives the additional resistance power to bacteria which makes them to not only tolerate harsh conditions but also resistant to antibiotics which lead to the emergence of bad bugs infections like multi drug resistant, extensively drug resistant and totally drug resistant bacteria.

77. **Biochip contains –**

(a) RNA

(b) DNA

(c) RNA and DNA

(d) RNA, DNA and Protein

U.P.P.C.S. (Mains) 2008

Ans. (d)

The biochip is a microchip designed or intended to function in a biological environment specially inside a living organism. These are likely to have an increasing impact on genetic diagnostics, drug discovery and basic research applications. In addition to DNA and RNA based chips, protein chips are also being developed with increasing frequency.

78. **Genico Technology is –**

(a) Defence system for prevention from AIDS

(b) A method for the development of species for food crops

(c) Technique for pre-information regarding genetic diseases

(d) Technique for prevention of cataract

U.P.P.C.S. (Pre) 1999

Ans. (c)

Genico Technology is a technique for pre-information regarding genetic diseases such as prenatal investigation regarding the diseases in the fetus.

79. **'Aerial metagenomics' best refers to which one of the following situations?**

(a) Collecting DNA samples from air in a habitat at one go

(b) Understanding the genetic makeup of avian species of a habitat

(c) Using air-borne devices to collect blood samples from moving animals

(d) Sending drones to inaccessible areas to collect plant and animal samples from land surfaces and water bodies.

I.A.S. (Pre) 2023

Ans. (a)

Metagenomics is the study of the structure and function of entire nucleotide sequences isolated and analyzed from all the organisms (typically microbes) in a bulk sample. Metagenomics is often used to study a specific community of microorganisms, such as those residing on human skin, in the soil or in a water sample. Aerial metagenomics typically refers to the study of genetic material (such as DNA or RNA) collected from the air, usually in the form of airborne particles or aerosols. It involves analysing the microbial communities present in the air and characterising their genetic diversity.

80. **Which one of the following approaches comes under the category of Biotechnology ?**

(a) Use of living organisms or substances obtained from them in industrial processes

(b) Modernizing the commercial industries to produce products for use in biological research

(c) Use of modern technology in research of biological mutilation

(d) Use of industrial technology to increase the biosphere

I.A.S. (Pre) 1993

Ans. (a)

Industrial use of living organisms, substances obtained from them or their biological processes is known as Biotechnology. For example use of yeast cells for liquor production comes under Biotechnology.

81. With reference to the role of biofilters in Recirculating Aquaculture System, consider the following statements:

1. Biofilters provide waste treatment by removing uneaten fish feed.

2. Biofilters convert ammonia present in fish waste to nitrate.

3. Biofilters increase phosphorus as nutrient for fish in water.

How many of the statements given above are correct?

(a) Only one (b) Only two

(c) All three (d) None

I.A.S. (Pre) 2023

Ans. (b)

Recirculating Aquaculture Systems (RAS) are unique engineered ecosystems that minimize environmental perturbation by reducing nutrient pollution discharge. A biofilter system purify the water and remove or detoxify harmful waste products and uneaten feed. Biofilters use microorganisms, which are capable of degrading many compounds. Hence, statement 1 is correct.

RAS typically employ a biofilter to control ammonia levels produced as a byproduct of fish protein catabolism. There are four primary sources of nitrogenous wastes : urea, uric acid, and amino acid excreted by the fish, organic debris from dead and dying organisms, uneaten feed, and feces, and nitrogen gas from the atmosphere. The process of ammonia removal by a biological filter is called nitrification, and consists of the successive oxidation of ammonia to nitrite and finally to nitrate. Nitrosomonas (ammonia-oxidizing), Nitrospira, and Nitrobacter (nitrite-oxidizing) species are thought to be the primary nitrifiers present in RAS biofilters. So, statement 2 is also correct.

Statement 3 is incorrect as biofilters do not increase phosphorus as a nutrient for fish in the water. Biofilters are also used to remove phosphorus waste by-products generated by fish.

82. Which of the following is recently evolved in Genetic Engineering ?

(a) Gene analysis (b) Gene mapping

(c) Gene splicing (d Gene synthesis

Uttarakhand P.C.S. (Pre) 2006

Ans. (b)

Recently evolved technology in genetic engineering is Gene Mapping. It is a method used for determining the location of gene and relative distances between genes on a chromosome. The essence of all genome mapping is to place a collection of molecular markers onto their respective positions on the genome. Molecular markers come in all forms. Genes can be viewed as one special type of genetic markers in the construction of genome maps and mapped the same way as any other markers.

83. Which of the following is used in Genetic-Engineering?

(a) Plastid (b) Plasmid

(c) Mitochondria (d) Ribosome

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (b)

Bacteria may posses extranuclear genetic materials consisting of DNA. These extranuclear cytoplasmic carriers of genetic information between cells of the same species and even between different species and genera, are termed as plasmids. These are used in genetic engineering.

84. Two bacteria found to be very useful in genetic engineering experiments are _______.

(a) Nitrosomonas and Kliebsiella

(b) Escherichia coli and Agrobacterium

(c) Nitrobacter and Azotobacter

(d) Rhizobium and Diplococcus

Uttarakhand P.C.S. (Pre) 2021

Ans. (b)

The most important bacteria in genetic engineering of plants has been the Ti plasmid of soil bacterium, Agrobacterium tumefaciens and Escherichia coli. Agrobacterium is a useful tool for plant transformation because it can carry, transfer, and integrate a gene of interest into the plant genome. E.coli is one of the most important bacteria used in the field of biotechnology and genetic engineering.

85. With reference to bacteriophages, which statement/s is/ are correct?

1. Bacteriophages are viruses that infect bacteria.

2. Bacteriophages are used in genetic engineering.

Select the correct answer using the codes given below :

Codes :

(a) 1 only

(b) 2 only

(c) Both (1) and (2)

(d) Neither (1) nor (2)

U.P. R.O./A.R.O. (Pre) 2017

Ans. (c)

A bacteriophage is a virus that infects and replicates within bacteria and is used as a tool or vector in genetic engineering for transferring DNA fragment of one organism to another organism.

86. If a mouse of over eight times its size has been produced by human growth hormone gene, the technique involved is called –

(a) Hybridization (b) Genetic engineering
(c) Mutation breeding (d) Hormonal feeding

I.A.S. (Pre) 1993

Ans. (b)

Such a process/technique is known as genetic engineering. By this method, not only the size of a creature can be changed or the traits can be changed but also a completely new organism can be formed.

87. Given : 1. Blood cells
2. Bone cells
3. Hair strands
4. Saliva

Samples taken for DNA testing in criminal investigation can be –

(a) 1, 2 and 3 only (b) 1 and 4 only
(c) 2 and 3 only (d) 1, 2, 3 and 4

U.P.P.C.S. (Pre) 2008

Ans. (d)

Forensic DNA profiling (also called DNA testing) is a technique employed by forensic scientists to identify individuals by characteristics of their DNA. DNA profiling is generally used in paternity testing and criminal investigation. Samples taken for DNA testing in a criminal investigation can be blood cells, bone cells, hair strands, semen, and saliva etc.

88. Assertion (A) : 'DNA Fingerprinting' has become a powerful tool to establish paternity and identity of criminals in rape and assault cases.

Reason (R) : Trace evidence such as hairs, saliva and dried semen are adequate for DNA analysis.

Code :

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not a correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

I.A.S. (Pre) 2000

R.A.S./R.T.S. (Pre) 2013

Ans. (a)

DNA fingerprinting also called DNA typing, DNA profiling, genetic fingerprinting, genotyping or identity testing in genetics, is a method of isolating and identifying variable elements within the base-pair sequence of DNA (deoxyribonucleic acid). The technique was developed by British geneticist Alec Jeffreys after he noticed that certain sequences of highly variable DNA (known as mini-satellites), which do not contribute to the functions of genes are repeated within genes. Jeffreys recognized that each individual has a unique pattern of mini-satellites (the only exceptions being multiple individuals from a single zygote, such as identical twins). DNA fingerprinting is a powerful tool to establish paternity and identity of criminals in rape and assault cases because DNA from trace evidence such as hairs, saliva and dried semen are adequate for DNA analysis.

89. The basis of DNA fingerprinting is :

(a) The double helix
(b) Errors in base sequence
(c) DNA replication
(d) DNA polymorphism

R.A.S/R.T.S (Pre) 2018

Ans. (d)

DNA fingerprinting is a form of identification based on sequencing specific non-coding portions of DNA that are known to have a high degree of variability from person to person. Most of our DNA is identical to each other, but there are inherited regions of our DNA that can vary from person to person. These variations are termed as polymorphisms and this forms the basis of DNA fingerprinting. Since DNA from every tissue (skin, blood, saliva, bone, hair-follicle, sperm etc.) from an individual show the same degree of polymorphism, they become very useful identification tool in forensic applications. Further, as the polymorphisms are inheritable from parents to children, DNA fingerprinting is the basis of paternity testing.

90. Which one of the following techniques can be used to establish the paternity of a child ?

(a) Protein analysis
(b) Chromosome counting
(c) Quantitative analysis of DNA
(d) DNA fingerprinting

U.P. U.D.A./L.D.A. (Spl.) (Pre) 2010

I.A.S. (Pre) 1997

Ans. (d)

See the explanation of above question.

91. **Which one of the given is useful for proving paternity?**
(a) Gene therapy
(b) Gene cloning
(c) D.N.A. Recombinant technology
(d) D.N.A. fingerprinting
U.P. Lower Sub. (Pre) 2002

Ans. (d)

See the explanation of above question.

92. **The latest technique used to establish identity of a human being based on biotechnological principle is–**
(a) Biomatrix analysis
(b) Genome sequencing
(c) DNA fingerprinting
(d) Karyotyping
R.A.S./R.T.S.(Pre) 2012

Ans. (c)

See the explanation of above question.

93. **DNA fingerprinting is used in the following area(s) :**
(a) Forensic cases
(b) Paternity dispute
(c) Conservation of endangered living being
(d) All of the above
U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (d)

See the explanation of above question.

94. **The first crime ever solved using the DNA fingerprinting technique was in England in the year–**
(a) 1963 (b) 1973
(c) 1983 (d) 1993
U.P.P.C.S. (Mains) 2006

Ans. (c)

DNA fingerprinting was first used forensically in the solving of the murder of two teenagers who had been raped and murdered in 1983 and 1986 in England.

95. **In addition to fingerprint scanning, which of the following can be used in the biometric identification of a person?**
1. Iris scanning
2. Retinal scanning
3. Voice recognition
Select the correct answer using the code given below :
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
I.A.S. (Pre) 2014

Ans. (d)

Biometrics identifiers are the distinctive, measurable characteristics used to label and describe individuals. Examples include fingerprint, face recognition, DNA, palm print, hand geometry, iris recognition, retinal recognition, gait, voice recognition etc. Thus iris scanning, retinal scanning and voice recognition all three can be used in the biometric identification of a person.

96. **The powder used for developing fingerprints on a multi-coloured surface is –**
(a) Gold dust
(b) Manganese dioxide
(c) Charcoal
(d) Fluorescent powder
U.P.P.C.S. (Mains) 2013

Ans. (d)

A new class of organic base fluorescent powders can be used for developing fingerprints on nonporous surfaces (e.g. plastics, polybags, glass, metals, etc.) and multicoloured glossy surfaces.

97. **Which one of the following is not achieved by transgenics?**
(a) Production of biodegradable plastic
(b) Production of edible vaccines
(c) Production of cloned animals
(d) Production of transgenic crops
U.P.P.C.S. (Pre) 2000

Ans. (c)

In biology, cloning is the process of producing similar populations of genetically identical individuals that occur in nature when organisms such as bacteria, insects or plants reproduce asexually.Transgenesis is the process of introducing an exogenous gene – called a transgene – into a living organism so that the organism will exhibit a new property and transmit that property to its offspring. Transgenic bacteria are used to produce antibiotics on an industrial scale, new protein drugs and to metabolize petroleum products or plastics for cleaning up the environment. Transgenic animals are useful in basic research for determining gene function.

98. **Change in the base sequence within the gene is called :**
(a) Breeding (b) Cloning
(c) Mutation (d) Fusion
U.P. P.C.S. (Pre) 2016

Ans. (c)